

Machine Technical Manual: High-Speed Automated IC Tester & Pick-and-Place Sorter

Domain Intelligence & Anomaly Diagnostics Specification for Model ATE-001

Document Identifier:	VAI-MAN-ATE-001	Document Version:	v1.0
Prototype Machine ID:	ATE-001	Generation Date:	2026-08-26
Process Stage:	Final Electrical Test & Sort	Document Status:	Synthetic Prototype Technical Manual
Target Platform:	VectorAI Predictive Engine	Knowledge Role:	Primary Domain Source

SYNTHETIC PROTOTYPE TECHNICAL MANUAL — VECTORAI SYSTEM SPECIFICATION

This document is artificially generated specifically for the VectorAI predictive maintenance demonstration, knowledge base retrieval (RAG), formula-based RUL calculation, and automated diagnostics.

The specifications, thresholds, sensor values, service-life hours, maintenance intervals, and operating parameters are synthetic and must NOT be used for real-world physical industrial equipment operation or maintenance.

Table of Contents / Knowledge Structure:

1. Document Information & Disclaimer

2. Machine Overview & Architecture

3. Major Subsystems & Components

4. Sensors & Telemetry Configuration

5. Sensor Threshold Definitions

6. Standard Operating Conditions

7. Maintenance Intervals & Service Life
8. Degradation Indicators & Physics

9. Deterministic Formula-Based RUL Model

10. Troubleshooting & Failure Symptoms

11. Diagnostic Knowledge & Failure Scenarios

12. Standard Maintenance Procedures

13. Document Revision History

14. VectorAI RAG Integration Notes

1. Document Information & Governance

This technical document provides authoritative machine specifications for **High-Speed Automated IC Tester & Pick-and-Place Sorter** (Identifier: ATE-001, Manual ID: VAI-MAN-ATE-001). It establishes the single source of truth for the VectorAI platform, defining telemetry sensor schemas, threshold limits, maintenance intervals, and deterministic degradation parameters.

Governance Principle: All diagnostic algorithms, RUL models, and threshold engines in VectorAI must strictly adhere to the limits and parameters defined herein. No machine learning regressions or stochastic models are permitted in the primary RUL calculation pipeline.

2. Machine Overview & Manufacturing Process

Process Stage: Final Electrical Test & Sort

Automated Test Equipment (ATE) system integrated with a high-throughput pick-and-place device handler, tri-temperature environmental conditioning chamber, and multi-bin sorter designed to execute parametric and functional electrical validation at speed, sorting verified ICs into pass/fail bins or tape-and-reel carriers.

Detailed Process Flow:

Packaged integrated circuits are fed from input tubes, trays, or bulk bowls. The high-speed robotic handler picks devices, brings them to temperature in an environmental chamber (-40°C to +125°C), and compresses them into precision pogo-pin test sockets on the Device-Under-Test (DUT) interface board. The ATE instrumentation executes DC parametric, AC timing, and functional RF/digital tests in milliseconds. Tested ICs are sorted into designated bin categories or transferred to high-speed tape-and-reel packaging.

Key Machine Subsystems:

- Automated High-Speed Pick-and-Place Device Handler & Shuttle
- Precision DUT Test Socket Interface & Pogo Pin Array
- Tri-Temperature Environmental Conditioning Chamber
- High-Speed Parametric Measurement Unit (PMU) & Pin Electronics
- 2D DataMatrix Optical Inspection & Laser Marking Station
- Multi-Bin Sorting Shuttles & Tape-and-Reel Packaging Output

3. Major Components & Degradation Characteristics

Component	Primary Function	Important Parameters	Degradation Indicators
DUT Interface Socket & Pogo Pin Block	Provides temporary, highly reliable electrical contact to IC package pins/balls with minimal contact resistance and signal distortion.	Pogo pin contact resistance (mΩ), compression stroke force (N), contact lifecycle count, socket surface temperature (°C).	Contact resistance elevation (>85 mΩ), false open-circuit test failures, pogo pin spring fatigue, socket debris buildup.
High-Speed Handler Robotic Carriage	Transfers untested chips to test sockets and sorts tested devices into pass/fail bin positions at accelerations up to 15G.	Carriage linear vibration (mm/s), positioning repeatability (μm), vacuum pickup seal integrity.	Handler vibration spikes (>0.90 mm/s), mechanical positioning jitter, device dropping, track belt slack.
Socket Actuator Press	Applies programmable compression force to seat device pins firmly into pogo pin matrix during test execution.	Actuator compression force (N), descent velocity (mm/s), force planarity balance.	Actuator force creep (>75 N), lead deformation on packages, pneumatic seal leakage.
Tri-Temp Environmental Chamber	Maintains precise device testing temperature from cold test (-40°C) to room temp (25°C) to hot burn-in (125°C).	Chamber air temperature (°C), thermal airflow rate (CFM), thermocouple accuracy.	Chamber temperature drift (>35°C in room test), fan bearing noise, thermal gradient across DUT sites.
Sorting Shuttle & Tape-and-Reel Unit	Directs tested ICs into up to 8 bin classes and seals passing units into carrier tape with heated cover tape shoe.	Sorting accuracy (μm), sealing shoe temperature (°C), tape peel force (N).	Sorting positioning error (>15 μm), bin mis-sorting, carrier tape jam, sealing shoe temperature decay.

4. Sensor Configuration & Telemetry Mapping

The following sensor schema represents the continuous telemetry parameters streamed by the machine controller into the VectorAI Edge Collector. Direction indicates whether increasing or decreasing values signify degradation.

Sensor ID	Sensor Name	Unit	Normal Range	Warning Range	Critical Range	Direction
vibration_handler	Handler Vibration	mm/s	0.2 – 0.6	0.6 – 0.9	0.9 – 2.5	HIGHER IS WORSE
temperature_chamber	Test Chamber Temp	°C	22.0 – 28.0	28.0 – 35.0	35.0 – 150.0	HIGHER IS WORSE
load_actuator	Socket Actuator Load	N	40.0 – 60.0	60.0 – 75.0	75.0 – 120.0	HIGHER IS WORSE
contact_resistance	Pogo Pin Contact Resistance	mΩ	25.0 – 55.0	55.0 – 85.0	85.0 – 200.0	HIGHER IS WORSE
sorting_accuracy	Bin Sorting Positional Error	μm	0.0 – 8.0	8.0 – 15.0	15.0 – 40.0	HIGHER IS WORSE
test_socket_temperature	Socket Surface Temp	°C	24.0 – 32.0	32.0 – 40.0	40.0 – 80.0	HIGHER IS WORSE

5. Sensor Threshold Definitions & Operational Boundaries

Sensor / Metric	Normal Operational Band	Warning Threshold Band	Critical Fault Band
Handler Vibration vibration_handler	0.2 to 0.6 mm/s Smooth handler carriage traverse and pick-and-place indexing.	0.6 to 0.9 mm/s Linear guide rail dry friction or belt tension relaxation.	0.9 to 2.5 mm/s Severe mechanical guide binding or loose carriage bearings causing device jams.
Test Chamber Temp temperature_chamber	22.0 to 28.0 °C Stable room-temperature parametric testing thermal envelope.	28.0 to 35.0 °C Chamber air circulation fan degradation or cleanroom thermal drift.	35.0 to 150.0 °C Thermal condition out of specification causing parametric test yield fallout.
Socket Actuator Load load_actuator	40.0 to 60.0 N Nominal compression seating pogo pins to optimal working travel.	60.0 to 75.0 N Actuator friction increase or socket guide pin resistance.	75.0 to 120.0 N Excessive force causing package lead bending or pogo pin destruction.
Pogo Pin Contact Resistance contact_resistance	25.0 to 55.0 mΩ Clean gold/palladium-alloy contact interfaces with minimal voltage drop.	55.0 to 85.0 mΩ Pogo pin tip oxidation or solder plating transfer buildup.	85.0 to 200.0 mΩ Severe contact degradation causing false electrical test failures (Yield crash).
Bin Sorting Positional Error sorting_accuracy	0.0 to 8.0 μm High-precision pocket placement into tape pocket or tray.	8.0 to 15.0 μm Linear belt stretch or optical encoder scale dust fouling.	15.0 to 40.0 μm High risk of device jamming on pocket walls and lead damage.
Socket Surface Temp test_socket_temperature	24.0 to 32.0 °C Optimal thermal baseline during high-frequency DUT switching.	32.0 to 40.0 °C Localized ohmic heating from elevated pin contact resistance.	40.0 to 80.0 °C Dangerous socket heating risking PEEK body softening and pin misregistration.

6. Standard Operating Conditions & Environmental Limits

Ambient Temperature:	20.0 - 23.0 °C (Cleanroom ISO Class 7)
Relative Humidity:	40% - 60% RH non-condensing (ESD safe environment)
Normal Operating Temperature:	25.0 °C (Room temp test) / -40.0 to +125.0 °C (Tri-Temp modes)
Operating Pressure:	0.50 - 0.60 MPa (CDA pneumatic supply)
Operating Speed / Velocity:	8,000 - 15,000 Units Per Hour (UPH)
Standard Cycle Time:	0.24 - 0.45 seconds per tested device
Production Schedule:	20 - 24 hours/day continuous
Recommended Operating Conditions:	Automated pogo pin cleaning brush cycle every 5,000 devices; socket life check every 250,000 cycles.
Max Continuous Operation:	120 hours before required socket electrical calibration and alignment verification

7. Maintenance Schedule & Expected Service Life

Component	Interval (Hours)	Expected Life	Required Maintenance Action & Procedure
DUT Pogo Pin Test Socket Block	100 hrs	1500 hrs	Run automated abrasive brush cycle, clean pin tips with isopropyl alcohol swab, check contact resistance. <i>Procedure:</i> Use Kelvin 4-wire micro-ohmmeter on gold test coupon. Replace socket when average resistance > 60 mΩ.
Handler Robotic Linear Drive Belts & Guides	500 hrs	6000 hrs	Clean linear rails, lubricate guide bearings with cleanroom synthetic grease, check belt tension with sonic meter. <i>Procedure:</i> Verify timing belt tension is 85 ± 5 Hz using optical/acoustic tension meter. Inspect belt teeth for wear.
Socket Actuator Pneumatic Press	350 hrs	4000 hrs	Inspect actuator guide bushings, replace pneumatic cylinder seals, calibrate compression load cell. <i>Procedure:</i> Place load cell calibration tool into socket nest; verify actuator delivers exactly 50 ± 2 N.
Tri-Temp Thermal Circulation System	750 hrs	8000 hrs	Clean air circulation blower fans, check thermoelectric Peltier modules, calibrate RTD temperature sensors. <i>Procedure:</i> Place 4-wire platinum RTD logger into chamber; verify thermal uniformity across all 4 test sites ± 0.5°C.
Tape-and-Reel Sealing Shoe & Cutter	250 hrs	3000 hrs	Clean Teflon sealing shoe surface, check heater cartridge, perform cover tape peel force test. <i>Procedure:</i> Run peel force tester on 500 mm of sealed carrier tape (verify peel force is 30 - 80 gf per EIA-481 standard).

8. Degradation Indicators & Physics of Failure

Each parameter below reflects physical mechanical or electrical wear in machine subsystems. The VectorAI anomaly engine monitors these indicators to calculate Remaining Useful Life (RUL).

Parameter	Normal Baseline	Degraded Condition	Critical Limit	Underlying Physics / Mechanism
Pogo Pin Contact Resistance	Average pin resistance between 25 and 55 mΩ with low variance.	Contact resistance climbs to 55 - 85 mΩ due to solder transfer and tip oxidation.	Resistance exceeds 85 mΩ; causes false parametric test bin failures and yield crashes.	Direct physical indication of solder intermetallic transfer, noble metal coating wear, or internal pogo spring fatigue.
Handler Carriage Vibration	Vibration amplitude 0.20 - 0.60 mm/s during high-speed indexing.	Vibration climbs to 0.60 - 0.90 mm/s due to guide rail dry friction or belt slack.	Vibration exceeds 0.90 mm/s; mechanical resonance causes device pickup misregistration.	Indicates mechanical guide wear, linear motor bearing play, or timing belt tooth wear.
Socket Actuator Load	Compression force maintains nominal 40 - 60 N.	Force climbs to 60 - 75 N as actuator friction increases.	Force exceeds 75 N; high risk of crushing delicate IC package leads.	Indicates pneumatic guide cylinder seal friction, mechanical guide binding, or load cell drift.
Test Chamber Temperature	Maintained strictly at 22 - 28 °C during room temp validation.	Temperature drifts to 28 - 35 °C.	Temperature exceeds 35 °C; thermal drift alters semiconductor bandgap parameters.	Indicates blower fan failure, thermal insulation loss, or environmental chamber control loop failure.
Bin Sorting Positional Error	Pocket placement offset below 8 μm.	Positional error increases to 8 - 15 μm.	Positional error exceeds 15 μm; device edges catch on pocket walls causing jams.	Indicates timing belt elongation, optical encoder scale contamination, or theta rotational backlash.

9. Deterministic Formula-Based RUL Model Configuration

Base Useful Life (BUL): 6000 operating hours

Mathematical Formula: VectorAI computes RUL deterministically without neural networks or stochastic regression. For each parameter *i*:

- For HIGHER_IS_WORSE: $d_i = (currentValue - healthyLimit) / (criticalLimit - healthyLimit)$
- For LOWER_IS_WORSE: $d_i = (healthyLimit - currentValue) / (healthyLimit - criticalLimit)$
- Clamp: $d_i = \max(0.0, \min(1.0, d_i))$
- Overall Degradation: $D_{total} = \sum (w_i \times d_i)$ [where $\sum w_i = 1.00$]
- Remaining Useful Life: $RUL = \max(0, BaseUsefulLife \times (1.0 - D_{total}))$

Parameter	Sensor ID	Unit	Weight (w_i)	Healthy Limit	Critical Limit	Direction
Contact Resistance	contact_resistance	mΩ	0.30	55.0	110.0	HIGHER IS WORSE

Handler Vibration	vibration_handler	mm/s	0.25	0.6	1.3	HIGHER IS WORSE
Actuator Load	load_actuator	N	0.20	60.0	90.0	HIGHER IS WORSE
Chamber Temperature	temperature_chamber	°C	0.15	28.0	45.0	HIGHER IS WORSE
Sorting Positional Error	sorting_accuracy	µm	0.10	8.0	22.0	HIGHER IS WORSE
TOTAL WEIGHT SUM	-	-	1.00 (100%)	-	-	VERIFIED OK

10. Troubleshooting & Failure Symptoms Table

The following troubleshooting matrix provides rapid diagnostic mappings between observable anomalies and root causes.

Symptom & ID	Severity	Related Sensors	Possible Causes	Recommended Corrective Action
Pogo Pin Contact Resistance Elevation / False Open Fails SYM-ATE-01	Critical	contact_resistance , test_socket_temperature	- Solder and tin plating transfer onto gold-plated pogo pin crown tips - Pogo pin internal helical spring fatigue or contact micro-oxidation - Dust and resin particulate accumulation inside socket guide holes	Run automated diamond cleaning brush sequence. If resistance > 60 mΩ persists, replace pogo pin socket insert.
Handler Carriage Vibration Surge / Pick Jitter SYM-ATE-02	High	vibration_handler, sorting_accuracy	- Linear guide rail dry friction and cleanroom grease breakdown - Timing belt tension relaxation or tooth pitch wear - Pick-and-place head mounting screw looseness	Clean linear guides, lubricate with cleanroom grease, adjust timing belt tension to 85 Hz with sonic meter.
Socket Actuator High Compression Force / Lead Bending SYM-ATE-03	High	load_actuator	- Pneumatic cylinder guide rod mechanical binding - Actuator downward stroke limit stop set too deep - Foreign particle trapped between device nest and socket face	Inspect socket nest for debris, verify actuator load calibration (50 N), lubricate cylinder guide pins.
Test Chamber Temperature Drift / Thermal Yield Shift SYM-ATE-04	Medium	temperature_chamber	- Environmental chamber air circulation fan bearing wear or failure - Thermoelectric Peltier module thermal paste degradation - Chamber thermal door seal gasket air leak	Inspect circulation blower, check Peltier drive current, verify 25.0°C chamber temperature stability.
Tape-and-Reel Pocket Placement Jam / Mis-Sort SYM-ATE-05	High	sorting_accuracy	- Sorting shuttle X/Y positioning belt stretch or encoder skip - Carrier tape sprocket indexing pin wear - Vacuum pickup collet tip wear causing device tilt during placement	Re-teach tape pocket pickup coordinates, replace vacuum collet, clean optical encoder strip.

11. Comprehensive Diagnostic Knowledge Base (Failure Scenarios)

This section contains 13 detailed synthetic failure scenarios used as the initial primary diagnostic knowledge base by the VectorAI Machine Agent. When an anomaly is detected, the agent queries this manual first before fallback to RAG.

Scenario 1: Pogo Pin Solder Transfer & Contact Resistance Spike (SCEN-ATE-001)	Severity: CRITICAL
Telemetry Sensor Pattern:	contact_resistance surges from 38 mΩ to 92 mΩ; false Continuity Fail rate climbs to 4.2%
Possible Root Causes:	Lead-free tin plating transferring from device balls to pogo crown tips Internal spring relaxation after 200,000 mechanical actuations Socket guide plate pocket contamination
Recommended Corrective Action:	Execute 20 strokes with abrasive cleaning coupon. Replace socket insert if resistance remains > 60 mΩ.
Verification & Recovery Steps:	Run automated socket diagnostic self-test with gold coupon > Verify pin resistance < 40 mΩ across all channels
Scenario 2: Handler Carriage Timing Belt Tooth Wear & Vibration (SCEN-ATE-002)	Severity: HIGH
Telemetry Sensor Pattern:	vibration_handler rises from 0.42 to 0.94 mm/s; sorting_accuracy error jumps to 16.5 μm
Possible Root Causes:	Kevlar timing belt tooth shear wear after 2,000 operating hours Belt tension dropped below 60 Hz Linear guide bearing ball recirculation fatigue
Recommended Corrective Action:	Replace timing belt, adjust tension to 85 Hz using acoustic meter, clean and re-grease linear rails.
Verification & Recovery Steps:	Execute high-speed multi-point positioning test > Verify repeatability < 5.0 μm
Scenario 3: Socket Actuator Cylinder Friction Binding (SCEN-ATE-003)	Severity: HIGH
Telemetry Sensor Pattern:	load_actuator climbs from 50 N to 78 N; cycle time delay alert
Possible Root Causes:	Pneumatic cylinder piston rod seal friction and contamination Socket guide bushing misaligned Force sensor amplifier baseline drift
Recommended Corrective Action:	Replace pneumatic actuator seal kit. Re-align socket guide pins. Calibrate force sensor.
Verification & Recovery Steps:	Perform 50 actuation cycles with load cell tool > Verify load is 50 ± 2 N consistently
Scenario 4: Test Chamber Air Blower Fan Bearing Failure (SCEN-ATE-004)	Severity: MEDIUM
Telemetry Sensor Pattern:	temperature_chamber rises from 24.5°C to 36.8°C; thermal gradient alert
Possible Root Causes:	Circulation fan brushless motor bearing seizure Peltier cooling heat sink fins clogged with dust Chamber intake air filter restriction
Recommended Corrective Action:	Replace circulation blower fan. Clean heat sink radiator fins. Replace intake air filter.
Verification & Recovery Steps:	Log chamber temperature over 20 minutes > Verify temperature stays at 25.0 ± 0.5°C

Scenario 5: Tape-and-Reel Sealing Shoe Temperature Decay (SCEN-ATE-005)	Severity: HIGH
Telemetry Sensor Pattern:	Cover tape peel force drops below 20 gf (carrier open defect risk)
Possible Root Causes:	Sealing shoe heater cartridge resistance increase Teflon tape build-up on sealing shoe blade Thermocouple probe loose
Recommended Corrective Action:	Clean sealing shoe with brass brush, replace heater element if resistance > 120 Ω , verify 160°C setpoint.
Verification & Recovery Steps:	Run peel force test on 1 meter of sealed tape (target 45 $\pm$ 10 gf)
Scenario 6: Vacuum Pickup Collet Tip Contamination and Device Drop (SCEN-ATE-006)	Severity: HIGH
Telemetry Sensor Pattern:	Handler pickup error rate climbs to 1.8%; vibration_handler nominal
Possible Root Causes:	Silicone suction cup rim coated with package mold compound dust Vacuum solenoid valve response delayed Vacuum generator venturi clogged
Recommended Corrective Action:	Clean suction cup tips with IPA swab. Inspect vacuum level (min -75 kPa).
Verification & Recovery Steps:	Run 200 dummy devices in loop mode > Verify zero drop errors
Scenario 7: 2D DataMatrix Laser Reader Optical Blur (SCEN-ATE-007)	Severity: MEDIUM
Telemetry Sensor Pattern:	Vision scan timeout increases by 15 ms per device; occasional unreadable ID
Possible Root Causes:	Optical scanner window coated with cleanroom dust film Laser scanner focal distance drifted due to bracket vibration LED illumination ring half-channel failure
Recommended Corrective Action:	Clean scanner optical window with lens tissue. Recalibrate focal distance and illumination.
Verification & Recovery Steps:	Test scan 100 sample 2D barcodes > Verify 100% first-pass read rate
Scenario 8: Pogo Pin Breakage Causing Short Circuit on DUT Site (SCEN-ATE-008)	Severity: CRITICAL
Telemetry Sensor Pattern:	contact_resistance on Channel 4 reads 0.0 m Ω (short) or infinite (open)
Possible Root Causes:	Mechanical fatigue fracture of internal pin spring Foreign metallic burr bridging adjacent socket pins Socket guide plate cracked
Recommended Corrective Action:	Remove socket block immediately. Replace broken pogo pin using pin insertion tool.
Verification & Recovery Steps:	Inspect all pins with 50x microscope > Perform pin-to-pin isolation test (> 100 M Ω)
Scenario 9: Handler Theta Rotation Backlash & Misalignment (SCEN-ATE-009)	Severity: MEDIUM
Telemetry Sensor Pattern:	sorting_accuracy theta error increases to $\pm$ 1.2 degrees
Possible Root Causes:	Harmonic drive theta gearbox wear Theta stepper motor coupling set screw loose Rotary optical home sensor dirty
Recommended Corrective Action:	Tighten motor shaft coupling set screw with threadlocker. Clean home optical sensor.
Verification & Recovery Steps:	Run theta rotation calibration routine > Verify angular repeatability < 0.1 deg

Scenario 10: Electrostatic Discharge (ESD) Grounding Resistance Drift (SCEN-ATE-010)	Severity: CRITICAL
Telemetry Sensor Pattern:	ESD grounding monitor alert; resistance climbs from 0.8 Ω to 15.4 Ω
Possible Root Causes:	Handler wrist strap grounding wire braid broken Conductive table mat ground snap corroded Socket grounding bus connection loose
Recommended Corrective Action:	Replace ESD grounding cable. Clean ground lug interface with wire brush. Verify resistance < 1.0 Ω .
Verification & Recovery Steps:	Measure point-to-ground resistance with calibrated megohmmeter (< 1.0 Ω)
Scenario 11: Carrier Tape Sprocket Hole Indexer Tooth Wear (SCEN-ATE-011)	Severity: HIGH
Telemetry Sensor Pattern:	Tape feeder jam alert; sorting_accuracy in tape direction drifts $\pm 20 \mu\text{m}$
Possible Root Causes:	Sprocket wheel teeth worn down from continuous plastic tape indexing Feeder advance stepper belt loose Cover tape peel tension too high pulling carrier backward
Recommended Corrective Action:	Replace indexing sprocket wheel. Calibrate tape advance pitch to $4.00 \pm 0.02 \text{ mm}$.
Verification & Recovery Steps:	Advance 100 tape pockets > Verify pocket center alignment with vision camera
Scenario 12: DUT Test Head Docking Latch Play (SCEN-ATE-012)	Severity: HIGH
Telemetry Sensor Pattern:	contact_resistance values fluctuate across all pins when handler vibrates
Possible Root Causes:	Pneumatic docking latch mechanism air pressure low Docking alignment guide pins worn Test head stiffener frame clamping torque uneven
Recommended Corrective Action:	Inspect docking latch air pressure (min 0.55 MPa). Replace worn docking guide pins.
Verification & Recovery Steps:	Execute docking repeatable contact resistance test > Verify stability < $\pm 2 \text{ m}\Omega$
Scenario 13: Socket Elastomer Body Localized Overheating (SCEN-ATE-013)	Severity: HIGH
Telemetry Sensor Pattern:	test_socket_temperature climbs from 28°C to 44.5°C during continuous DC stress test
Possible Root Causes:	High current power pin pogo resistance elevated causing I^2R ohmic heating Socket thermal cooling airflow obstructed Continuous high duty cycle testing without dwell time
Recommended Corrective Action:	Replace high-current pogo pins with heavy-duty crown pins. Clean socket bottom cooling channels.
Verification & Recovery Steps:	Thermal camera inspection of socket under full test load > Verify max temp < 32°C

12. Standard Operating & Maintenance Procedures (SOP)

SOP-01: Shift Start Operational Inspection & Tool Touch-Off

1. Verify Clean Dry Air (CDA) supply pressure is within 0.55 - 0.65 MPa and dew point is < -40°C.
2. Ensure cleanroom ambient temperature is stabilized within specified operating window (20 - 24°C).
3. Run automated sensor zero-calibration and optical target homing sequence.
4. Process 2 calibration test pieces; inspect output dimensions and telemetry stability on VectorAI Dashboard.

SOP-02: Tool Replacement & Calibration Protocol

1. Lock out and tag out (LOTO) machine controller before entering mechanical enclosure.
2. Use calibrated preset torque wrench to fasten tool fasteners to exact manufacturer specification.
3. Execute optical non-contact tool alignment and measure runout / contact resistance.
4. Log maintenance record in VectorAI with work order ID, technician ID, and timestamp.

SOP-03: Anomaly Escalation Protocol

When a sensor crosses from Warning into Critical threshold band:

1. VectorAI generates automated Severity: High/Critical Incident.
2. System automatically references Section 11 Diagnostic Knowledge for recommended immediate actions.
3. If Remaining Useful Life (RUL) falls below 48 hours, dynamic production rerouting is triggered to alternate healthy lines.

13. Document Revision History

Revision	Release Date	Author / Organization	Description of Changes
v1.0	2026-08-26	VectorAI Knowledge Engineering	Initial synthetic technical specification and diagnostic manual release.
v0.9-RC	2026-08-15	VectorAI Systems Lab	Draft sensor thresholds, RUL weight configuration, and failure scenarios.

14. VectorAI RAG Integration & Knowledge Indexing

This document is formatted with distinct semantic headings, tables, and structured diagnostic scenarios to facilitate high-precision chunking and vector embedding in the VectorAI RAG pipeline.

- **Chunk Association:** Every indexed chunk retains metadata: machineId, manualId, section.
- **Scoped Retrieval:** Diagnostics for machine ATE-001 prioritize chunks where machineId == 'ATE-001'.